

Q1 - 24 June - Shift 1

The number of points where the function

$$f(x) = \begin{cases} |2x^2 - 3x - 7| & \text{if } x \leq -1 \\ [4x^2 - 1] & \text{if } -1 < x < 1 \\ |x + 1| + |x - 2| & \text{if } x \geq 1 \end{cases}$$

[t] denotes the greatest integer  $\leq t$ , is

discontinuous is \_\_\_\_\_.

Space for your notes:

Q2 - 24 June - Shift 2

$$\text{Let } f(x) = \begin{cases} \frac{\sin(x - [x])}{x - [x]}, & x \in (-2, -1) \\ \max\{2x, 3[x]\}, & |x| < 1 \\ 1, & \text{otherwise} \end{cases}$$

Space for your notes:

where [t] denotes greatest integer  $\leq t$ . If m is the number of points where f is not continuous and n is the number of points where f is not differentiable, then the ordered pair (m, n) is :

- (A) (3, 3) (B) (2, 4)  
(C) (2, 3) (D) (3, 4)

Q3 - 25 June - Shift 2

## Questions

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Let  $f(x) = [2x^2 + 1]$  and  $g(x) = \begin{cases} 2x - 3, & x < 0 \\ 2x + 3, & x \geq 0 \end{cases}$ , where  $[t]$  is the greatest integer  $\leq t$ . Then, in the open interval  $(-1, 1)$ , the number of points where  $f \circ g$  is discontinuous is equal to \_\_\_\_\_

Space for your notes:

## Q4 - 26 June - Shift 1

Let  $f, g : \mathbb{R} \rightarrow \mathbb{R}$  be two real valued functions

defined as  $f(x) = \begin{cases} -|x + 3|, & x < 0 \\ e^x, & x \geq 0 \end{cases}$  and

$g(x) = \begin{cases} x^2 + k_1x, & x < 0 \\ 4x + k_2, & x \geq 0 \end{cases}$ , where  $k_1$  and  $k_2$  are

real constants. If  $(g \circ f)$  is differentiable at  $x = 0$ , then  $(g \circ f)(-4) + (g \circ f)(4)$  is equal to :

- (A)  $4(e^4 + 1)$  (B)  $2(2e^4 + 1)$   
(C)  $4e^4$  (D)  $2(2e^4 - 1)$

Space for your notes:

## Q5 - 26 June - Shift 2

Let  $f(x) = \min \{1, 1 + x \sin x\}$ ,  $0 \leq x \leq 2\pi$ . If  $m$  is the number of points, where  $f$  is not differentiable and  $n$  is the number of points, where  $f$  is not continuous, then the ordered pair  $(m, n)$  is equal to

- (A)  $(2, 0)$  (B)  $(1, 0)$   
(C)  $(1, 1)$  (D)  $(2, 1)$

Space for your notes:

## Q6 - 28 June - Shift 1

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## Questions

MathonGo

Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be defined as

$$f(x) = \begin{cases} [e^x], & x < 0 \\ ae^x + [x-1], & 0 \leq x < 1 \\ b + [\sin(\pi x)], & 1 \leq x < 2 \\ [e^{-x}] - c, & x \geq 2 \end{cases}$$

where  $a, b, c \in \mathbb{R}$  and  $[t]$  denotes greatest integer less than or equal to  $t$ . Then, which of the following statements is true?

(A) There exists  $a, b, c \in \mathbb{R}$  such that  $f$  is continuous of  $\mathbb{R}$ .

(B) If  $f$  is discontinuous at exactly one point, then

$$a + b + c = 1.$$

(C) If  $f$  is discontinuous at exactly one point, then

$$a + b + c \neq 1.$$

(D)  $f$  is discontinuous at atleast two points, for any values of  $a, b$  and  $c$ .

Space for your notes:

**Q7 - 28 June - Shift 2**

Let  $f, g: \mathbb{R} \rightarrow \mathbb{R}$  be functions defined by

$$f(x) = \begin{cases} [x], & x < 0 \\ |1-x|, & x \geq 0 \end{cases} \text{ and}$$

$$g(x) = \begin{cases} e^x - x, & x < 0 \\ (x-1)^2 - 1, & x \geq 0 \end{cases}$$

where  $[x]$  denote the greatest integer less than or equal to  $x$ . Then, the function  $f \circ g$  is discontinuous at exactly :

(A) one point (B) two points

(C) three points (D) four points

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Q8 - 29 June - Shift 1

Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be a function defined by :

$$f(x) = \begin{cases} \max_{t \leq x} \{t^3 - 3t\}; & x \leq 2 \\ x^2 + 2x - 6; & 2 < x < 3 \\ [x - 3] + 9; & 3 \leq x \leq 5 \\ 2x + 1; & x > 5 \end{cases}$$

Space for your notes:

Where  $[t]$  is the greatest integer less than or equal to  $t$ . Let  $m$  be the number of points where  $f$  is not differentiable and  $I = \int_{-2}^2 f(x) dx$ . Then the ordered pair  $(m, I)$  is equal to :

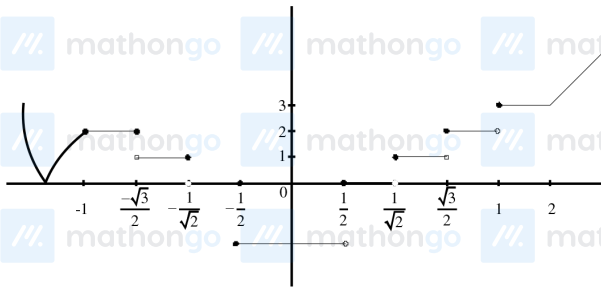
(A)  $\left(3, \frac{27}{4}\right)$  (B)  $\left(3, \frac{23}{4}\right)$

(C)  $\left(4, \frac{27}{4}\right)$  (D)  $\left(4, \frac{23}{4}\right)$

**Answer Key****Q1 (7)****Q2 (C)****Q3 (62)****Q4 (D)****Q5 (B)****Q6 (C)****Q7 (B)****Q8 (C)**

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Q1 (7)



Q2 (C)

$$f(x) = \begin{cases} \frac{\sin(x+2)}{x+2}, & x \in (-2, -1) \\ \max\{2x, 0\}, & x \in (-1, 1) \\ 1, & \text{otherwise} \end{cases}$$

$$f(-2^+) = \lim_{h \rightarrow 0} f(-2+h) = \lim_{h \rightarrow 0} \frac{\sinh}{h} = 1$$

$f$  is continuous at  $x = -2$

$$f(-1^-) = \lim_{h \rightarrow 0} \frac{\sin(-1-h+2)}{(-1-h+2)} = \sin 1$$

$$f(-1) = f(-1^+) = 0$$

$f(1^+) = 1$  &  $f(1^-) = 0 \Rightarrow f$  is not continuous at  $x = 1$

$f$  is continuous but not diff. at  $x = 0$

$$\Rightarrow f \text{ is discontinuous at } x = -1 \text{ \& } 1 \left. \begin{array}{l} m = 2 \\ \& f \text{ is not diff. at } x = -1, 0 \text{ \& } 1 \end{array} \right\} \Rightarrow n = 3$$

Q3 (62)

$$f(g(x)) = [2g^2(x)] + 1$$

$$= \begin{cases} [2(2x-3)^2] + 1; x < 0 \\ [2(2x+3)^2] + 1; x \geq 0 \end{cases}$$

$\therefore$  fog is discontinuous whenever  $2(2x-3)^2$  or

$2(2x+3)^2$  belongs to integer except  $x = 0$ .

$\therefore$  62 points of discontinuity.

**Q4 (D)**



$$f(x) = \begin{cases} x+3 & ; x < -3 \\ -(x+3) & ; -3 \leq x < 0 \\ e^x & ; x \geq 0 \end{cases}$$

$$g(x) = \begin{cases} x^2 + k_1 x & ; x < 0 \\ 4x + k_2 & ; x \geq 0 \end{cases}$$

$$g(f(x)) = \begin{cases} f(x)^2 + k_1 f(x) & ; f(x) < 0 \\ 4f(x) + k_2 & ; f(x) \geq 0 \end{cases}$$

$$g(f(x)) = \begin{cases} (x+3)^2 + k_1(x+3) & ; x < -3 \\ (x+3)^2 - k_1(x+3) & ; -3 \leq x < 0 \\ 4e^x + k_2 & ; x \geq 0 \end{cases}$$

check continuity at  $x = 0$

$$g \circ f(0) = g(f(0^-)) = g(f(0^+))$$

$$4 + k_2 = 9 - 3k_1 = 4 + k_2$$

$$3k_1 + k_2 = 5 \quad \dots(a)$$

differentiate

$$(g(f(x)))' = \begin{cases} 2(x+3) + k_1 & ; x < -3 \\ 2(x+3) - k_1 & ; -3 \leq x < 0 \\ 4e^x & ; x \geq 0 \end{cases}$$

$$6 - k_1 = 4$$

$$k_1 = 2 \quad \dots(b)$$

$$\therefore k_1 = 2, k_2 = -1$$

$$g \circ f(x) = \begin{cases} (x+3)^2 + 2(x+3) & ; x < -3 \\ (x+3)^2 - 2(x+3) & ; -3 \leq x < 0 \\ 4e^x - 1 & ; x \geq 0 \end{cases}$$

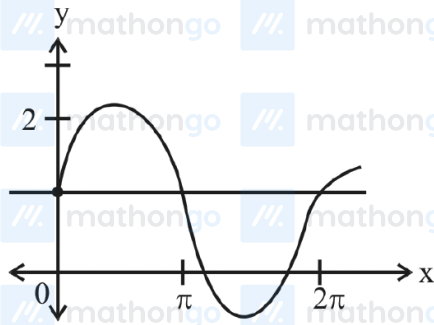
$$g \circ f(-4) + g \circ f(4) = 4e^4 - 2$$

$$\Rightarrow 2(2e^4 - 1)$$

Q5 (B)

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No. of non-differentiable points = 1 (m)

No. of not continuous points = 0 (n)

(m, n) = (1, 0)

**Q6 (C)**

$f(x)$  is discontinuous at  $x = 1$

For continuous at  $x = 0$ ;  $a = 1$

For continuous at  $x = 2$ ;  $b + c = 1$

$a + b + c = 2$

**Q7 (B)**

Check continuity at  $x = 0$  and also check

continuity at those  $x$  where  $g(x) = 0$

$g(x) = 0$  at  $x = 0, 2$

$\text{fog}(0^+) = -1$

$\text{fog}(0) = 0$

Hence, discontinuous at  $x = 0$

$\text{fog}(2^+) = 1$

$\text{fog}(2^-) = -1$

Hence, discontinuous at  $x = 2$

**Q8 (C)**

$$f(x) = \begin{cases} x^3 - 3x, & x \leq -1 \\ 2, & -1 < x < 2 \\ x^2 + 2x - 6, & 2 < x < 3 \\ 9, & 3 \leq x < 4 \\ 10, & 4 \leq x < 5 \\ 11, & x = 5 \\ 2x + 1, & x > 5 \end{cases}$$

Clearly  $f(x)$  is not differentiable at

$$x = 2, 3, 4, 5 \Rightarrow m = 4$$

$$I = \int_{-2}^{-1} (x^3 - 3x) dx + \int_{-1}^2 2 \cdot dx = \frac{27}{4}$$